

International University of Business Agriculture and Technology

Lab Report

Submitted By

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Project:

An Online E-Learning Platform

Project Description

The project is an online e-learning platform designed to provide an interactive, user-friendly and scalable environment for online education.
The system allows students, instructors, and administrators to collaborate in a digital learning ecosystem that offers access to courses, video lectures, quizzes similar to modern platforms like Coursera, Udemy, and edX.

This project aims to bridge the gap between learners and educators by offering flexible, self-paced learning opportunities accessible from anywhere.
Through this platform, instructors can create, manage, and publish courses while students can enroll, learn and take assessments.

The platform will be developed using the Agile (Scrum) model ensuring iterative development, user feedback integration and continuous improvement throughout the development process.

Project Objectives

- To develop a web-based e-learning platform that supports online courses, video content, and assessments.
- To ensure interactive learning experiences through discussion forums, progress tracking, and feedback.
- To provide role-based access (Admin, Instructor, Student) for secure and organized management.
- To create a responsive and scalable platform that can grow with user demand.
- To enable data-driven insights for instructors and administrators (e.g., student performance, course popularity).

System Users and Roles

Admin: Manages the entire platform — users, courses, categories, reports, and overall system maintenance.

Instructor: Creates, updates, and manages courses, quizzes, and student progress.

Student: Registers, enrolls in courses, watches videos, takes quizzes, and earns certificates.

Core Functionalities

1. User Management

- User registration and authentication (students, instructors, admins).
- Profile management (photo, bio, contact info, learning history).
- Role-based access control.

2. Course Management

- Course creation by instructors (title, description, syllabus, pricing, etc.).
- Upload and organize course materials (videos, documents, PDFs, slides).
- Course categorization and tagging for easy discovery.
- Course approval system (admin verification before publishing).

3. Enrollment & Learning

- Students can browse, search, and enroll in courses.
- Video streaming with playback controls.
- Progress tracking (percentage completed, last lesson viewed).
- Resume learning from where the user left off.
- Downloadable learning resources (if permitted).

4. Assessment & Certification

- Quizzes and assignments integrated into each course module.
- Automatic quiz grading and result display.

5. Communication & Interaction

- Discussion forums or Q&A under each course.
- Direct messaging between student and instructor.
- Course reviews and ratings.
- Notifications for course updates, deadlines, and new messages.

6. Payment and Monetization (Optional Advanced Feature)

- Secure online payment gateway (for paid courses).
- Instructor revenue tracking and payout system.
- Discounts and promotional offers.

7. Administration Panel

- Dashboard showing platform statistics (number of users, active courses, etc.).
- User and course management tools.
- Report generation (course performance, student activity).
- Feedback and issue tracking from users.

8. Analytics and Insights

- Track student engagement and performance trends.
- Course completion and dropout rates.
- Instructor dashboard showing course performance metrics.

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I have choose SCRUM from Agile process model to do our project.

Why SCRUM Model is chosen

1. Small, Cross-Functional Team (5 Members)

- Scrum is designed for small teams (typically 3–9 people).
- Our 5-person team can easily take on defined Scrum roles — *Product Owner*, *Scrum Master*, and *Developers*.
- This ensures clear responsibilities and smooth communication.

2. Frequent Requirement Changes

- E-learning platforms often evolve based on user feedback (teachers, students, administrators).
- Scrum's short development cycles (sprints) allow the team to adapt quickly to new requirements or design changes without disrupting the whole project.

3. Incremental and Iterative Delivery

- Instead of waiting until the end, Scrum delivers the website in increments — for example:

  Sprint 1: Login & registration

  Sprint 2: Course catalog & browsing

  Sprint 3: Video playback & quizzes

  Sprint 4: Progress tracking & analytics

- This helps stakeholders see tangible progress and test early features.

4. Better Collaboration and Communication

- Daily stand-up meetings and sprint reviews ensure everyone stays aligned.
- Problems are identified early, reducing misunderstandings and rework.
- The Scrum Master helps remove obstacles so the team stays productive.

5. High Product Quality

- Continuous testing, reviews, and sprint retrospectives help maintain quality throughout development.
- Bugs are detected early, not at the end, which reduces time and cost.

6. Stakeholder Involvement and Feedback

- Stakeholders (teachers, students, management) can review the product at the end of each sprint.
- Their feedback helps refine requirements and improve the system's usability.

7. Transparency and Control

- Scrum artifacts like the Product Backlog and Sprint Burndown Chart give a clear view of progress.
- The team and client can track what's done, what's in progress, and what's coming next.

8. Supports Continuous Improvement

- After every sprint, the team holds a retrospective meeting to discuss what worked well and what can improve.
- This makes the team more efficient over time which is a key Agile principle.

Complex Engineering Problems in E-Learning Site

1. Scalability & Performance
    - Handle thousands of concurrent learners.
    - Fast search, caching, and load balancing.
    - Efficient content delivery via CDNs.
2. Video Management
    - Upload, transcode, and stream videos at multiple qualities.

- Protect videos from piracy.

3. Security & Privacy

    - Secure logins, payments, and user data.
    - Prevent cheating or content theft.

4. Personalization & Recommendations

    - Suggest courses using ML on user behavior and content similarity.
    - Track and adapt learning paths.

5. Assessments & Progress Tracking

    - Auto-grading quizzes and assignments.
    - Track completion and sync progress across devices.
    - Safe code execution for programming courses.

6. Real-Time Interactions

    - Live classes, chat, Q&A, whiteboards using WebSockets or WebRTC.
    - Low-latency, high-availability architecture.

7. Analytics & Data Pipelines

    - Collect and analyze engagement data.

    - Build dashboards for instructors and admins.